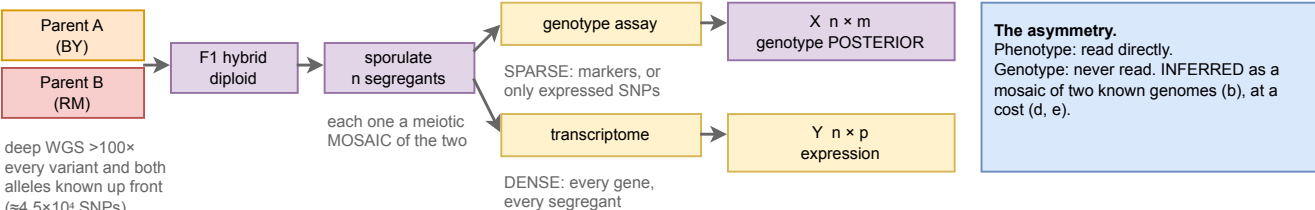
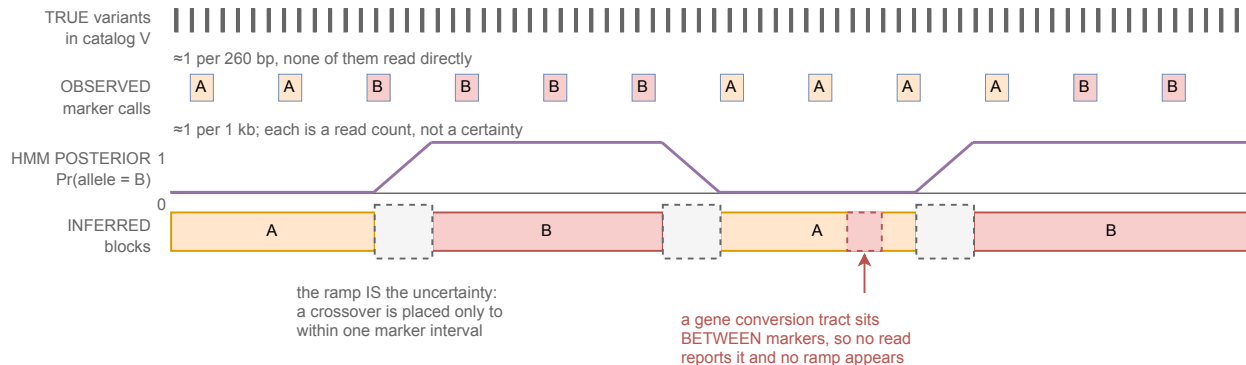


a The cross, and the two things measured on every segregant



b One segregant, one chromosome: from sparse reads to a posterior



Why sparse reads suffice.

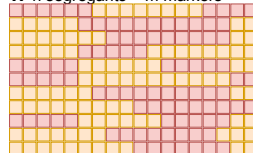
Each locus is ASSIGNED to one of two sequenced parents.
 ≈ 90 crossovers $\rightarrow \approx 10^2$ blocks; one marker fixes every variant in its block.

What no marker fixes.

Boundary intervals: $\approx 0.8\%$ of sites ambiguous.
 Gene conversion (≈ 46 per meiosis, ≈ 2 kb): invisible between markers.
 De novo mutation, aneuploidy, non-reference sequence: outside the model.

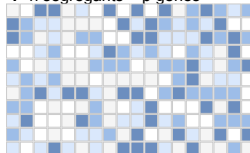
c What the two matrices look like

$X \ n \text{ segregants} \times m \text{ markers}$



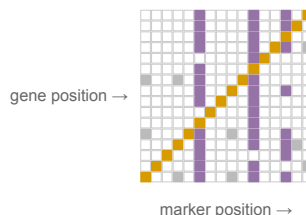
blocks, not noise: each row is a mosaic, so columns are linked

$Y \ n \text{ segregants} \times p \text{ genes}$



$p \approx 5,000$ to $6,000$; a VECTOR per strain, not a scalar

d The resulting map: cis on the diagonal, trans in bands



● cis: a variant acting on its OWN gene, usually in the promoter

● trans hotspot: one regulatory variant moving hundreds of transcripts

each cell is one significant (gene, marker) pair: a row of the QTL table

e What a record stores, as built for the Bloom 2019 segregant panels

Stored.

One record per segregant $\times$ condition.
 Genotype: SegregantGenotype, a sibling of Genotype. HaplotypeBlocks (chr, start, end, parent, posterior, n markers) between two pinned SegregantParents.
 Variant set: a derived view.

Not stored, and still open.

QTL table: an estimate, not a Phenotype.
 Intergenic variants: a mosaic needs no gene, a cis-eQTL still has none; per-variant attribution stays open.